

Exploratory Data Analysis (EDA) on Titanic Dataset

Task 5 – Data Analyst Internship

Name:

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Dataset:

Titanic Dataset (train.csv)

Tools Used:

- Python
 - Jupyter Notebook
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
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Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Titanic dataset to understand the data, identify patterns, discover relationships between variables, and extract meaningful insights using statistical methods and visualizations.

Dataset Description

The Titanic dataset contains information about passengers such as their age, gender, passenger class, fare, ticket details, cabin, and survival status. The dataset consists of **891 passenger records** and **12 columns**.

Data Exploration

The following steps were performed:

- Loaded the dataset using Pandas.
- Displayed the first few rows using head().
- Checked dataset shape and column names.

- Examined data types using `info()`.
 - Generated summary statistics using `describe()`.
 - Identified missing values using `isnull().sum()`.
 - Counted unique values using `nunique()`.
 - Checked survival counts using `value_counts()`.
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Visualizations Performed

The following visualizations were created:

- Age Distribution Histogram
 - Survival Count Plot
 - Passenger Class Distribution
 - Gender Distribution
 - Fare Distribution
 - Survival by Gender
 - Survival by Passenger Class
 - Age vs Survival Boxplot
 - Fare vs Survival Boxplot
 - Survival by Embarked Port
 - Correlation Heatmap
 - Pairplot
 - Survival Percentage Pie Chart
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Key Findings

- The dataset contains **891 passenger records**.
- Missing values were found in the **Age**, **Cabin**, and **Embarked** columns.
- Most passengers were between **20 and 40 years** of age.
- Male passengers were more numerous than female passengers.
- Most passengers traveled in **Third Class**.

- Female passengers had a significantly higher survival rate than male passengers.
 - First-class passengers had better survival chances than passengers in lower classes.
 - Passengers who paid higher fares were more likely to survive.
 - The correlation heatmap showed generally weak correlations among most numerical features.
 - Pairplot helped visualize relationships between age, fare, passenger class, and survival.
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Conclusion

Exploratory Data Analysis helped in understanding the Titanic dataset by identifying missing values, examining feature distributions, and discovering relationships between important variables. The analysis revealed that factors such as gender, passenger class, and ticket fare had a significant influence on passenger survival. EDA provides valuable insights that can support future predictive modeling and machine learning tasks.

Outcome

This project improved my understanding of:

- Data Exploration
 - Data Visualization
 - Statistical Analysis
 - Feature Relationships
 - Pattern Identification
 - Python Libraries for Data Analysis
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Save the document as:

Task5_Report.pdf